

Drona Agrawal

Email | LinkedIn

EDUCATION

National University of Singapore

2026 – 2030

Doctor of Philosophy in Chemical and Biomolecular Engineering

Birla Institute of Technology, Pilani, KK Birla Goa Campus

2022 – 2025

B.E. Chemical Engineering

CGPA: 9.14/10.0

Minor in Physics

TECHNICAL FAMILIARITY

Programming Languages: Python, MATLAB, Julia

Software: MATLAB, COMSOL, Aspen Plus

Areas: Control Theory, Machine Learning-based Optimization, Numerical Methods, Computational Fluid Dynamics

RESEARCH EXPERIENCE

BITS Pilani, KK Birla Goa Campus, India

Jan'26 – Present

Research Intern

Supervisors: Dr. Sundari R. (BITS) & Dr. Dipin Pillai (IIT-Kanpur)

- Developed Direct Numerical Simulation (DNS) Model of Surfactant-Laden (Insoluble) Two-Phase Stratified Flow under Electric Field to study perturbations using Level-Set Method.
- Validated the DNS model using deformation studies performed on Surfactant-Laden (Insoluble) Simple Droplet in a micro-channel.
- Validated the stratified flow model against Linear Stability Analysis.

Purdue University, West Lafayette, IN, USA

May'25 – Mar'26

Research Intern (Remote)

Supervisor: Dr. Can Li

- Worked on Optimization of Pooling Problem, a non-convex QCP, in collaboration with ExxonMobil.
- Developed a self-supervised learning framework to enable a low-fidelity parametric model to give good quality feasible solutions closer to the high-fidelity model.
- Tested available benchmark instances on our pipeline, achieving an overall ~5% average error while also generating feasible solutions.

TEACHING

Undergraduate Teaching Assistant

Jan'25 – May'25

CHE F242: Numerical Methods for Chemical Engineers

- Conducted MATLAB tutorials for the second-year course on Numerical Methods for a class of 104. Also assisted in developing problems for conducting problem-solving tutorials.

VOLUNTEERING AND LEADERSHIP

President

Sept'24 – Sept'25

Department of Chemical Engineering, BITS Pilani, KK Birla Goa Campus

- Elected unopposed with 100% votes to lead the department student body comprising 400+ students.
- Led initiatives to enhance faculty-student interaction. Helped reconstitute several department activities and functionalities that were affected during COVID.

Project Lead and Event Manager

Nov'22 – May'24

Nirmaan Organization, Student Chapter, BITS Pilani, Goa Campus

- Led and managed events to help 120+ students in Zari Slum, Goa, to excel in their education.
- Volunteered in various fundraising events to add to the cause.

AWARDS AND RECOGNITIONS

Emerging Leader in Chemical Engineering- Awarded by the Department of Chemical Engineering, BITS Pilani, KK Birla Goa Campus, to recognize my efforts and contributions to the department while serving as the President.